



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 3 Multiple Choice

9189/3

NOVEMBER 2008 SESSION

1 hour

Additional materials:

- Data Booklet
- Mathematical tables and/or calculator
- Multiple Choice answer sheet
- Soft clean eraser
- Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

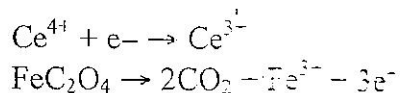
This question paper consists of 14 printed pages and 2 blank pages.

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Section A

For each question there are four possible answers, A, B, C and D. Choose the one you consider to be correct.

- 1 What is the volume of cerium (IV) sulphate of concentration 2.0 mol dm^{-3} required to oxidise 25.0 cm^3 of 0.6 mol dm^{-3} iron (II) ethanedioate solution given the following half-equations?



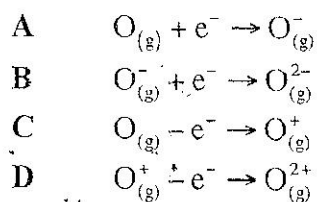
- A 25.0 cm^3
 B 22.5 cm^3
 C 7.5 cm^3
 D 2.5 cm^3
- 2 The first six ionisation energies of an element, Z, are as shown.

940 (first), 2100, 4140, 5820, 7170, 7950 kJ mol^{-1}

What is the formula of the chloride of Z?

- A ZCl
 B ZCl_2
 C ZCl_3
 D ZCl_4
- 3 Which substance does **not** have a tetrahedral shape?
- A silica
 B trichloromethane
 C trichloroethene
 D silicon

- 4 Which process is exothermic?



5

What is the correct order of increasing deviation of the given gases from ideal gas behaviour?

- A helium $\rightarrow$ hydrogen $\rightarrow$ nitrogen $\rightarrow$ ammonia
- B ammonia $\rightarrow$ nitrogen $\rightarrow$ hydrogen $\rightarrow$ helium
- C hydrogen $\rightarrow$ nitrogen $\rightarrow$ helium $\rightarrow$ ammonia
- D nitrogen $\rightarrow$ hydrogen $\rightarrow$ helium $\rightarrow$ ammonia

6

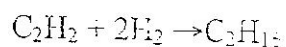
Which reaction involves the lattice energy of caesium fluoride?

- A $\text{Cs}_{(s)} + \frac{1}{2}\text{F}_{2(g)} \rightarrow \text{CsF}_{(s)}$
- B $\text{Cs}_{(s)} + \text{F}_{(g)} \rightarrow \text{CsF}_{(s)}$
- C $\text{Cs}_{(g)}^+ + \text{F}_{(g)}^- \rightarrow \text{CsF}_{(s)}$
- D $\text{Cs}_{(s)}^+ + \text{F}_{(g)}^- \rightarrow \text{CsF}_{(s)}$

7

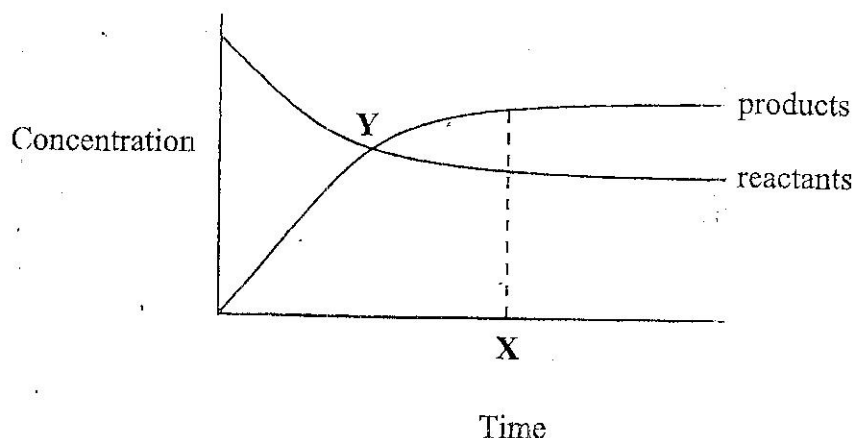
Heats of combustion of ethane, acetylene, and hydrogen are -1593 kJmol^{-1} , -1310 kJmol^{-1} and -285 kJmol^{-1} respectively.

What is the enthalpy change for the hydrogenation of acetylene?



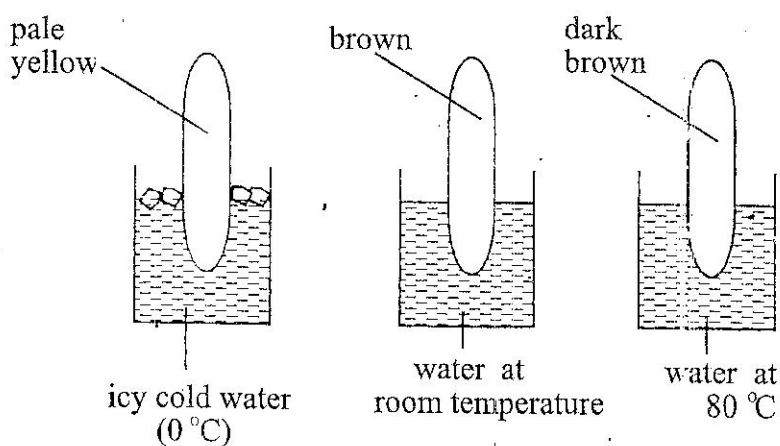
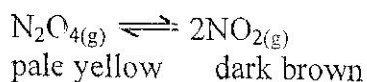
- A -2418 kJmol^{-1}
- B $+2418 \text{ kJmol}^{-1}$
- C $+487 \text{ kJmol}^{-1}$
- D -487 kJmol^{-1}

- 8 The diagram shows the variation in the concentration of reactants and products for a reversible reaction.



Which statement about the reaction is correct?

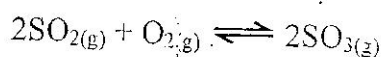
- A Concentrations of both reactants and products change most rapidly at point Y.
 - B The system reaches a state of dynamic equilibrium from time X.
 - C The chemical reaction ceases at X.
 - D The equilibrium constant is obtained at Y.
- 9 The diagram shows changes which occur to an equilibrium mixture of N_2O_4 and NO_2 at different temperatures. N_2O_4 decomposes to NO_2 according to the equation below.



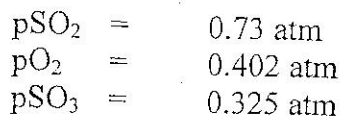
Which statement is correct?

- A Formation of N_2O_4 from NO_2 is endothermic.
- B K_c decreases in icy cold water.
- C K_c decreases in hot water.
- D At 0°C only $\text{N}_2\text{O}_{4(g)}$ is present.

- 10 Oxidation of sulphur dioxide is a reversible process.



The equilibrium partial pressures at 1000K were found to be as follows.



What is the value of K_p ?

- A 0.493 atm⁻¹
- B 0.903 atm⁻¹
- C 1.107 atm⁻¹
- D 2.028 atm⁻¹

- 11 Which statement is correct about the given reaction?



- A SO_4^{2-} is a conjugate base of HSO_4^- .
- B OH^- is a conjugate base of HSO_4^- .
- C H_2O is a stronger acid than HSO_4^- .
- D SO_4^{2-} is a stronger base than OH^- .

- 12 Which is the correct order of decreasing boiling points for the compounds $\text{CH}_3\text{CO}_2\text{CH}_3$, $\text{CH}_3\text{CH}_2\text{OH}$ and $\text{CH}_3\text{CH}_2\text{CO}_2\text{Na}$?

- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} > \text{CH}_3\text{CO}_2\text{CH}_3 > \text{CH}_3\text{CH}_2\text{CO}_2\text{Na}$
- B $\text{CH}_3\text{CH}_2\text{CO}_2\text{Na} > \text{CH}_3\text{CO}_2\text{CH}_3 > \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- C $\text{CH}_3\text{CO}_2\text{CH}_3 > \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} > \text{CH}_3\text{CH}_2\text{CO}_2\text{Na}$
- D $\text{CH}_3\text{CH}_2\text{CO}_2\text{Na} > \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} > \text{CH}_3\text{CO}_2\text{CH}_3$

- 13 Which statement about Group IV monoxides CO to SnO is correct?

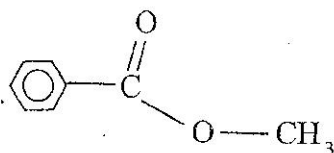
- A They are all good oxidising agents.
- B They become more acidic down the group.
- C They are unstable compared to the dioxides.
- D They are more acidic than the dioxides.

- 14 A certain element forms a chloride which is a volatile liquid that is readily hydrolysed by water to give a white precipitate and an acidic solution. What is the chloride of this element?

- A Al_2Cl_6
- B CCl_4
- C PbCl_4
- D PCl_5

- 15 Which statement is true for all of the compounds NaCl, NaBr and NaI?
- A They are oxidised by concentrated sulphuric acid to form corresponding halogens.
 - B They react with aqueous silver nitrate to form precipitates which dissolve in excess ammonia solution.
 - C They dissolve in water to form acidic solutions.
 - D They reduce fluorine to form sodium fluoride.
- 16 An oxide, X, reacts with aqueous sodium hydroxide to produce a salt Y. Y reacts with dilute hydrochloric acid to form a colourless gas.
- What is X?
- A SO_3
 - B Al_2O_3
 - C SnCl_2
 - D CO_2
- 17 The electronic configuration of a metal ion, X^{3+} , is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$.
- What is X?
- A Vanadium
 - B Chromium
 - C Manganese
 - D Iron
- 18 What is the oxidation state of platinum, Pt, in the compound below?
 $[\text{Pt}(\text{NH}_3)_4\text{SO}_4]^{2+} (\text{Br}^-)_2$
- A +1
 - B +2
 - C +3
 - D +4
- 19 Which list contains elements that all do not react with water?
- A magnesium, phosphorous, silicon
 - B sulphur, magnesium, phosphorous
 - C phosphorous, sulphur, silicon
 - D silicon, sulphur, magnesium
- 20 Which statement about both silicon dioxide and lead dioxide is correct?
- A PbO_2 is giant ionic whereas SiO_2 is simple molecular.
 - B PbO_2 is entirely basic whereas SiO_2 is acidic.
 - C PbO_2 is a better oxidising agent than SiO_2 .
 - D PbO_2 is more thermally stable than SiO_2 .

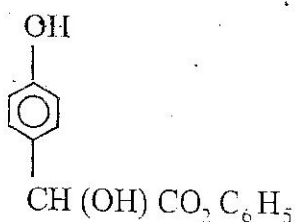
- 21 Which compound is formed when methylbenzoate is refluxed with dilute hydrochloric acid?



methylbenzoate

- A $\text{CH}_3\text{CO}_2\text{H}$
 B CH_3OCHO
 C $\text{C}_6\text{H}_5\text{CO}_2\text{H}$
 D $\text{C}_6\text{H}_5\text{COCl}$

- 22 Which statement about compound X, shown below is true?



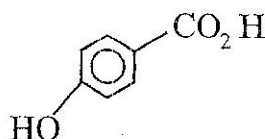
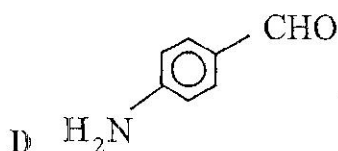
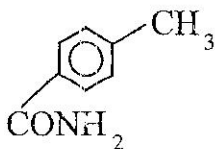
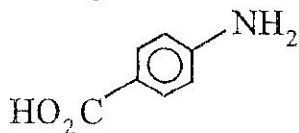
X

- A It is optically active.
 B It reacts with sodium carbonate.
 C It contains one aromatic ring.
 D It undergoes an electrophilic addition reaction with bromine.

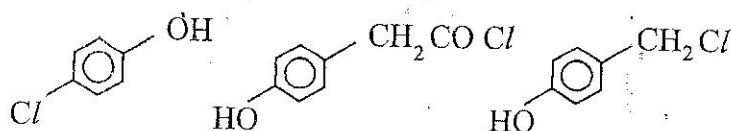
- 23 In which pair do both compounds react separately with dilute HCl and aqueous NaOH ?

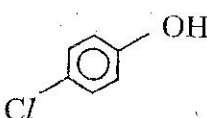
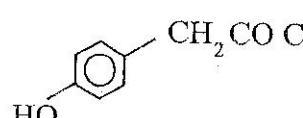
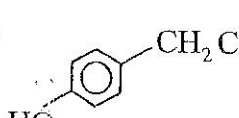
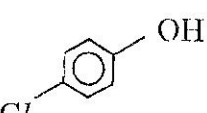
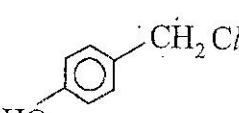
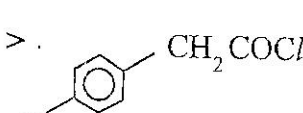
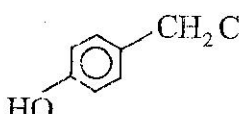
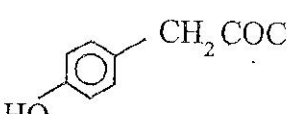
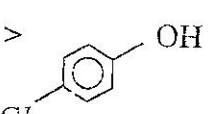
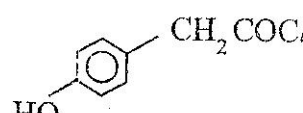
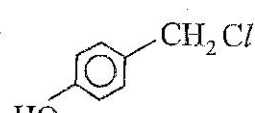
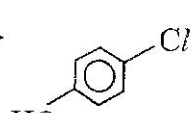
- A $\text{H}_2\text{NCH}_2\text{CH}_2\text{OH}$, $\text{CH}_3\text{CH}_2\text{CONH}_2$
 B $\text{CH}_3\text{CH}_2\text{CONH}_2$, $\text{CH}_3\text{CH}_2\text{COCH}_2\text{NH}_2$

C



- 24 Which is the correct order of decreasing reactivity of the given compounds with water?

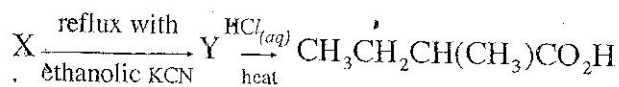


- A  >  > 
- B  >  > 
- C  >  > 
- D  >  > 

- 25 What type of reaction takes place between butanal and hydrogen cyanide?

- A electrophilic addition
B nucleophilic addition
C electrophilic substitution
D nucleophilic substitution

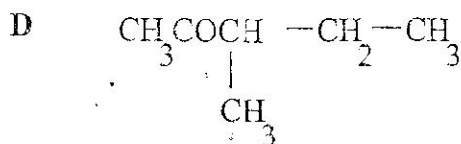
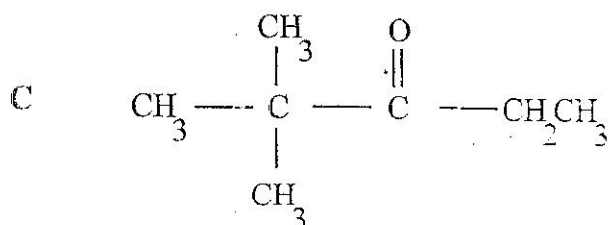
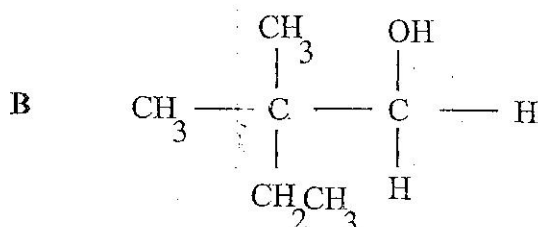
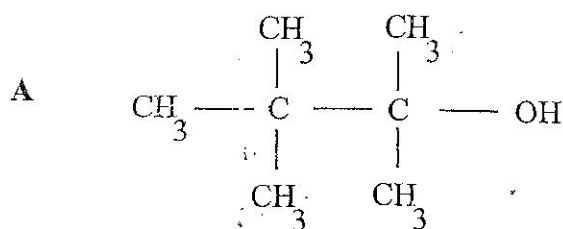
- 26 Substance X reacts as shown.



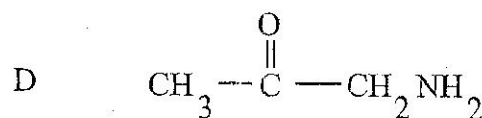
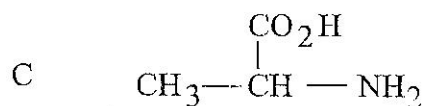
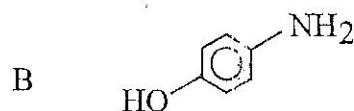
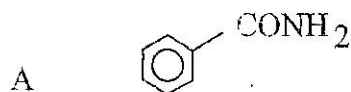
What is X?

- A $\text{CH}_3\text{CH}_2\text{COCH}_3$
B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$
C $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$
D $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CHO}$

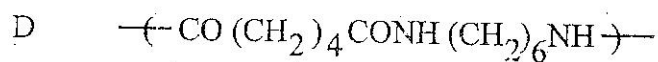
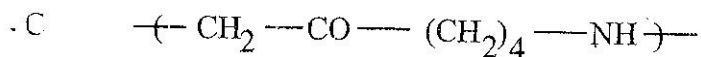
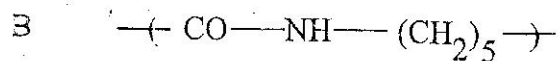
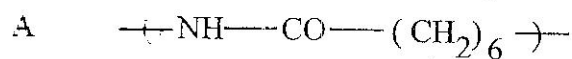
- 27 Which compound gives an orange precipitate when warmed with alkaline aqueous iodine solution?



- 28 Which substance reacts with aqueous sodium hydroxide to produce ammonia gas?



29 Which structure shows one repeat unit of nylon 6?



30 Which is the correct expression for pH of a buffer consisting of a weak base and its salt?

A $\text{pH} = -\log_{10} K_b + \log_{10} \frac{[\text{salt}]}{[\text{base}]}$

B $\text{pH} = -\log_{10} K_b + \log_{10} \frac{[\text{base}]}{[\text{salt}]}$

C $\text{pH} = 14 - \log_{10} K_b + \log_{10} \frac{[\text{base}]}{[\text{salt}]}$

D $\text{pH} = 14 + \log_{10} K_b + \log_{10} \frac{[\text{base}]}{[\text{salt}]}$

Section B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Which substance(s) contain(s) dative bonds?

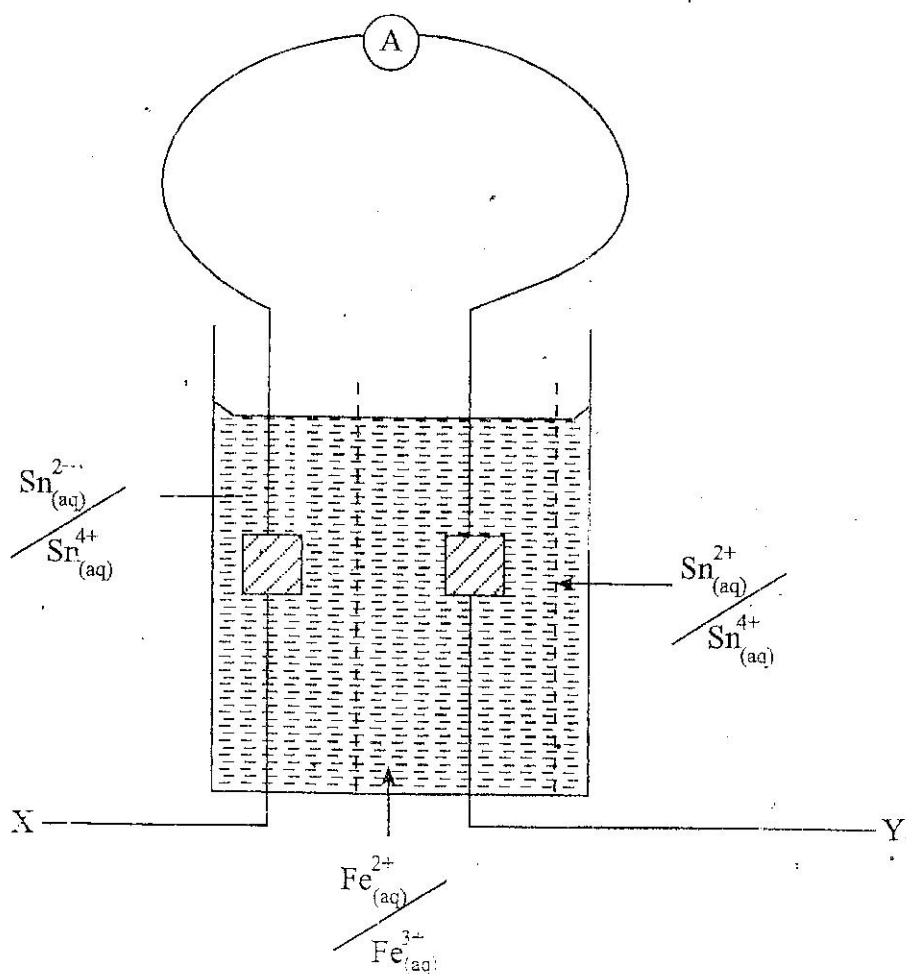
- 1 ammonium chloride
- 2 nitric acid
- 3 aqueous copper (II) ion

32 Which reaction(s) is/are endothermic?

- 1 $\text{Sr}_{(\text{g})}^{+} \rightarrow \text{Sr}_{(\text{g})}^{2+} + \text{e}^{-}$
- 2 $\text{F}_{(\text{g})} + \text{e}^{-} \rightarrow \text{F}_{(\text{g})}^{-}$
- 3 $\text{K}_{(\text{g})}^{+} + \text{e}^{-} \rightarrow \text{K}_{(\text{g})}$

Use of the Data Booklet is relevant to this question.

- 33 A cell is set up as shown in the diagram below.



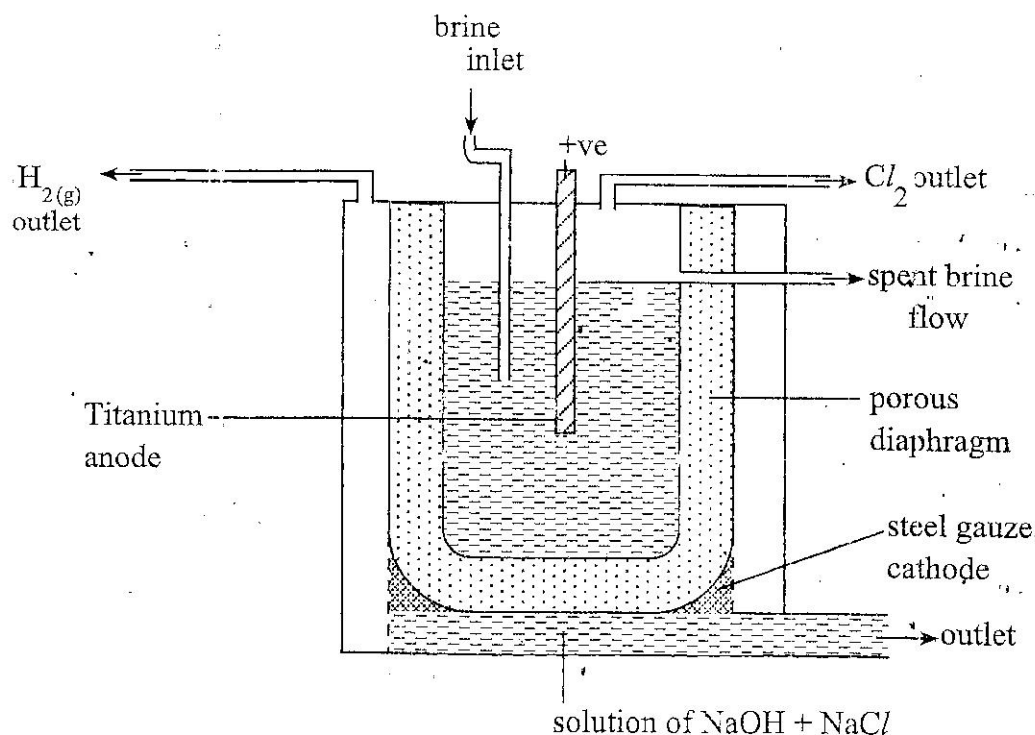
X and Y are platinum electrodes.

Which statement(s) is/are true about the cell?

- 1 Change taking place at X is $\text{Sn}^{2+} \rightarrow \text{Sn}^{4+} + 2\text{e}^-$.
 - 2 Change taking place at Y is $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$.
 - 3 $E^\circ_{\text{cell}} = +0.92 \text{ V}$.
- 34 Which of the following is/are true for a 0.1 mol dm^{-3} solution of aqueous ammonia, given that $K_b = 1.8 \times 10^{-5} \text{ mol dm}^{-3}$ at 298 K ?
- 1 $[\text{OH}^-] = 1.34 \times 10^{-3} \text{ mol dm}^{-3}$
 - 2 $\text{pH} = 11.13$
 - 3 $[\text{NH}_4^+] = 0.1 \text{ mol dm}^{-3}$

35

The diagram shows the diagram cell for manufacture of chlorine.



Which statement(s) is/are true about the process?

- 1 Volume of chlorine produced is twice volume of hydrogen.
- 2 Mixing of anode product and $\text{NaOH}_{(\text{aq})}$ results in disproportionation.
- 3 Anode product reacts with sodium thiosulphate to produce sodium sulphate.

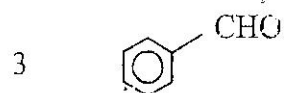
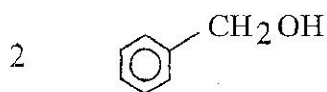
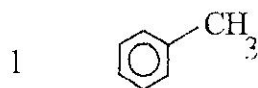
36

Which statement(s) about Group II compounds is/are correct?

- 1 Thermal stability of carbonates increases down the group.
- 2 pH of solutions obtained on dissolving the oxide in water decreases down the group.
- 3 Solubility of sulphates increases down the group.

37

Which substance(s) turn(s) sodium dichromate from orange to green?



38 Which substance(s) react(s) with both acid and alkali?

1 PbO_2

2 Al_2O_3

3 ZnO

39 When heated, 3-hydroxy-propanoic acid reacts as follows.



Which reagent(s) can be used to distinguish between the two acids?

1 Br_2

2 KMnO_4

3 PCl_5

40 Which statement(s) is/are true about the compound $\text{C}_6\text{H}_5\text{CHCH}_2$?

1 It reacts with steam.

2 It shows cis-trans isomerism.

3 It is optically active.